Uniform Trace-Class Möbius–Bergman Families with Quantified Order Sensitivity

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24 August 2026

Abstract

We promote a two-branch Möbius transfer operator from one digit pair to the rectangle $(a,b) \in [3,7/2] \times [6,7]$. The closed branch images remain separated by at least $1/45$, while the unweighted composition sum on normalized Bergman space is trace class with norm at most $89/16$ and is trace-norm Lipschitz in both parameters. Every word has an exact quadratic-surds trace, giving all power traces and a primitive Fredholm product. The non-cyclic same-count words *aaabb* and *aabab* have matrix trace gap $a(b-a)^2 \geq 175/8$, so order sensitivity holds uniformly. These are source-dynamical statements, not an arithmetic or target spectral identification.

1 Family-level progress

Let $\mathbb{D} = \{z : |z| < 1\}$ and equip $A^2(\mathbb{D})$ with normalized area, so $e_n(z) = \sqrt{n+1}z^n$ is orthonormal. Freeze

$$\phi_x(z) = \frac{1}{x+z}, \qquad \mathcal{L}_{a,b} = C_{\phi_a} + C_{\phi_b}, \qquad (a,b) \in \mathcal{R}_* = [3,7/2] \times [6,7].$$

The advance over a fixed pair is uniform: no conclusion below depends on choosing one parameter sample.

2 Uniform geometry and nuclearity

The image of $\overline{\mathbb{D}}$ under ϕ_x is the closed disk with center $x/(x^2-1)$ and radius $1/(x^2-1)$. Its real extremes are $1/(x+1)$ and $1/(x-1)$, hence the gap between the two images is

$$g(a,b) = \frac{1}{a+1} - \frac{1}{b-1} \geq \frac{1}{9/2} - \frac{1}{5} = \frac{1}{45}.$$

This is a rectangle theorem rather than a corner sample: $1/(a+1)$ decreases with a, whereas $-1/(b-1)$ increases with b. The minimum is therefore attained at the exact corner $(a,b) = (7/2, 6)$.

Theorem 1 (uniform trace class)**.** *For every $(a,b) \in \mathcal{R}_*$, $\mathcal{L}_{a,b}$ is trace class and $\|\mathcal{L}_{a,b}\|_1 \leq 89/16$.*

Proof. The rank-one expansion $C_{\phi_x} f = \sum_{n\geq 0} \langle f, e_n \rangle \sqrt{n+1}\phi_x^n$ is nuclear because $\sup_{\mathbb{D}} |\phi_a| \leq 1/2$ and $\sup_{\mathbb{D}} |\phi_b| \leq 1/5$. Using $\sqrt{n+1} \leq n+1$ gives

$$\|\mathcal{L}_{a,b}\|_1 \leq \sum_{n\geq 0}(n+1)2^{-n} + \sum_{n\geq 0}(n+1)5^{-n} = 4 + \frac{25}{16}.$$

□

Proposition 1 (trace-norm continuity)**.** *For two parameters in $\mathcal{R}_*$,*

$$\|\mathcal{L}_{a,b} - \mathcal{L}_{a',b'}\|_1 \le 4|a-a'| + \frac{5}{32}|b-b'|.$$

Proof. The resolvent identity gives $|\phi_x - \phi_y| \le |x-y|/(m-1)^2$ on $\mathbb{D}$ when $x, y \ge m$. If both symbols have modulus at most r, then $|\phi_x^n - \phi_y^n| \le nr^{n-1}|\phi_x - \phi_y|$. The same rank-one decomposition as in Theorem 1 now bounds the trace norm by

$$\frac{|x-y|}{(m-1)^2}\sum_{n\ge 1}(n+1)nr^{n-1} = \frac{2|x-y|}{(m-1)^2(1-r)^3}.$$

The choices $(m, r) = (3, 1/2)$ and $(6, 1/5)$ give 4 and 5/32. □

3 All words and the determinant

Put $M_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & x \end{smallmatrix}\right)$. For $M_w = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, write $t = \operatorname{tr} M_w$, $\delta = (-1)^{|w|}$, and $\Delta = t^2 - 4\delta$. Directly solving the fixed equation gives

$$z_w = \frac{A - D + \sqrt{\Delta}}{2C}, \quad \lambda_w = \frac{t - \sqrt{\Delta}}{t + \sqrt{\Delta}}, \quad \operatorname{Tr} C_{\Phi_w} = \frac{1}{2} + \frac{t}{2\sqrt{\Delta}}.$$

Here $M_{x_1} \cdots M_{x_n}$ represents $\phi_{x_1} \circ \cdots \circ \phi_{x_n}$. Composition operators multiply in the reversed function order. Reversal is a bijection on all length-n words, so it preserves the complete trace sum without asserting a false termwise order equality. Expansion of $\mathcal{L}_{a,b}^n$ therefore gives every power trace. Regrouping repetitions in the Fredholm logarithm yields

$$\det(I - z\mathcal{L}_{a,b}) = \prod_{[p]}\prod_{k\ge 0}(1 - z^{|p|}\lambda_p^k), \qquad |z| < \tfrac{1}{2}. \tag{1}$$

The left side is entire by Theorem 1; equation (1) states raw-factor convergence only in the displayed disk.

4 Uniform order sensitivity

The words *aaabb* and *aabab* have the same symbol population and are not cyclic rotations. Exact multiplication gives

$$\begin{aligned} t_1 &= a^3b^2 + a^3 + 2a^2b + 2ab^2 + 3a + 2b, \\ t_2 &= a^3b^2 + 4a^2b + ab^2 + 3a + 2b, \end{aligned}$$

so

$$t_1 - t_2 = a(b-a)^2 \ge \frac{175}{8}.$$

Also $t_1 \le 10731/4$. For $F(t) = 1/2 + t/(2\sqrt{t^2+4})$, $F'(t) = 2/(t^2+4)^{3/2}$. The mean-value theorem therefore gives

$$F(t_1) - F(t_2) \ge \frac{2800}{(10731^2 + 64)^{3/2}} > 0.$$

5 Exact receipt and boundary

At nine rational sentinels, the release reconstructs all words through period ten: $9(2^1+\cdots+2^{10}) = 18{,}414$ rooted receipts and $9 \cdot 226 = 2{,}034$ primitive-parameter receipts. A separate symbolic run performs 18,379 exact checks; replay is byte-identical and 40 repaired-hash plus one stale-hash mutations are rejected. The grid audits the implementation and is not the proof of uniformity.

The exact verdict is

$$(\mathrm{A1\_WEAK}, \mathrm{A2\_FAIL}, \mathrm{A3\_FAIL}, \mathrm{A4\_FAIL}).$$

There is no target divisor, prime or zero table, arithmetic Euler factor, root number, automorphy claim, unitary lift, or Hilbert–Pólya operator. Route B is unauthorized under `NO_BAD_EULER_OR_ROOT_NUMBER`.

6 Conclusion

C137 proves that nonlinear order sensitivity and trace-class determinant control persist on a compact parameter family. Connecting this source-owned family to any external target is a separate unsupported problem.